SOC Investigation Report

SL-IR-2026-001 | Suspicious Authentication Activity | 14 October 2026

Case	Environment	Assessment
b54a0961 | High severity	Training Lab (Synthetic)	Simulated brute-force pattern; compromise unconfirmed

1. Executive summary

A synthetic authentication sequence shows 12 failed logins to the admin account on 192.168.1.10 from 185.220.101.5, followed by a successful login from the same source. Two later outbound TCP/443 connections from a different internal IP, 192.168.1.105, targeted 185.220.101.5. SentinelLab generated a high-severity brute-force alert and a critical IOC match against an active internally configured indicator. These observations support investigation of a simulated attack pattern but do not establish unauthorized access, command-and-control traffic, or exfiltration.

2. Scope and methodology

Evidence basis: original SentinelLab JSON and CSV event exports (64 records each), the evidence manifest, the extracted authentication timeline, and analyst-observed screenshots of case, alert, event, and threat-intelligence views. Review consisted of event-sequence reconstruction, field-level correlation, IOC configuration review, detection validation, and AI-output quality assessment. No live attack or active containment was performed.

3. Evidence inventory and integrity

Item	Size	SHA-256 (full)
sentinellab-events.json	63,349 bytes	64e5c58dc4afafe41f41690d8f7a6fbcc5d602a313ea9cbaf0d23f8a0533217f
sentinellab-events.csv	22,018 bytes	f332141d27fa8257c89b0ca421126052913f5c777ef6777952b908b6285505f4

Hashes were recomputed from the available original exports and match the pre-existing evidence manifest. The files are application exports, not original endpoint log files. The SentinelLab interface displayed shortened hash prefixes for attached evidence; a full custody audit trail has not been independently verified.

4. Dataset composition

Authentication	Network	File	Process	Total
27	15	12	10	64

All 64 JSON records are flagged synthetic. Matching JSON/CSV event-ID sets were previously checked; complete field-by-field equivalence is not asserted.

5. Reconstructed event timeline (IST)

Time, 10 Oct 2026	Observed event	Analyst interpretation
23:06:34–23:09:52	12 failed admin authentications from 185.220.101.5 to 192.168.1.10	Pattern consistent with brute-force attempts
23:11:34	Successful admin authentication, same source and target	Authorization unknown
23:20:34	Outbound TCP/443: 192.168.1.105 → 185.220.101.5	Temporal/IP association only
23:21:34	Second outbound TCP/443: same endpoints	No process or payload attribution

All timestamps above are local IST, converted from UTC timestamps in the exported records.

6. Detection and threat-intelligence validation

Brute Force Authentication alert e095b20e: High severity, 80% application-reported confidence, 12 supporting failed-authentication events, MITRE ATT&CK T1110. The later successful login is outside that 12-event support set.

Known Malicious IOC Match alert 871d008f: Critical severity, 90% application-reported confidence, 64 matched events, with 50 shown in the supporting-events panel. The active IP indicator 185.220.101.5 was listed under Internal TI, added 10 October 2026, with tags c2 and tor-exit. The tag labels and confidence basis were not independently validated. Matching source or destination IP fields across different synthetic records does not imply 64 distinct malicious actions.

MITRE mapping is based on the application display: T1110 for the brute-force alert; T1071 was displayed on the IOC alert. T1071 is not independently demonstrated by application-protocol telemetry or verified command-and-control content.

7. Network correlation findings

Proxy event b19df32e-dba7-425a-9b28-5518299fe2b1 recorded TCP connection success from 192.168.1.105 to 185.220.101.5:443, on proxy host 10.0.0.45, tagged outbound-suspicious. Its process, username and source port fields were null. The authentication target 192.168.1.10 differs from the proxy connection source 192.168.1.105. No reliable causal linkage or host pivot was demonstrated.

8. AI explanation validation

The AI identified the repeated failed logins and suggested generally sensible defensive controls. However, it speculated about absent MFA/lockout controls, IP reputation, and a UDP/22 event without enough supporting context, and it did not adequately explain the selected IOC alert provenance or the successful-login/network relationship. AI output was treated as a triage aid, not as evidence.

9. Conclusions and outstanding questions

• Confirmed: synthetic failed-login burst, subsequent successful login, two outbound TCP/443 events, and matches to an internally configured IOC.

• Unconfirmed: whether the login was authorized, whether distinct internal hosts were causally connected, and whether IOC labels reflect independently verified intelligence.

• Not established: real-world compromise, command-and-control sessions, malware execution, lateral movement, or data exfiltration.

• Case remains Investigating because validation tasks are open and the application does not support completing its static task entries.

10. Recommended next steps

• In a controlled lab, verify successful-login authorization and host ownership using independent identity and endpoint evidence.

• Enrich proxy events with process/endpoint context, DNS/TLS metadata and connection telemetry if available.

• Record IOC feed provenance, first/last seen, confidence method and validation date.

• Preserve original exports and hashes; document evidence access and transfer events in a full custody log.

• Improve alert scope and AI grounding before relying on automated incident disposition.

11. Case-study integrity statement

This report describes a controlled synthetic SOC investigation and SentinelLab functional validation. No real victim, attacker attribution, confirmed intrusion, or verified external threat-intelligence enrichment is claimed.